HDOC Day 44

Aim

To write a menu-driven C program using separate user-defined functions to reverse each word in a sentence without changing word order and to print the initials of a name.

Algorithm

1. Start the program and display a menu with three choices.

2. For choice 1, read a sentence, locate each word and reverse its characters in place while retaining the word order.

3. For choice 2, read a name and print the uppercase first letter of each word followed by a period.

4. Repeat the menu until the user selects Exit.

5. Stop.

C Program

```
#include <stdio.h>
#include <string.h>
#include <ctype.h>

#define SIZE 500

void readLine(char str[], int size) {
    if (fgets(str, size, stdin) != NULL)
        str[strcspn(str, "\n")] = '\0';
}

void reverseWords(char str[]) {
    int i = 0, start, end;
    while (str[i] != '\0') {
        if (isspace((unsigned char)str[i])) {
            i++;
            continue;
        }
        start = i;
        while (str[i] && !isspace((unsigned char)str[i]))
            i++;
        end = i - 1;
        while (start < end) {
            char temp = str[start];
            str[start] = str[end];
            str[end] = temp;
            start++;
            end--;
        }
    }
}

void printInitials(const char name[]) {
    int i, newWord = 1, found = 0;
    for (i = 0; name[i] != '\0'; i++) {
        if (isspace((unsigned char)name[i])) {
            newWord = 1;
        } else if (newWord) {
            printf("%c.", toupper((unsigned char)name[i]));
```

```
            newWord = 0;
            found = 1;
        }
    }
    if (!found) printf("No name entered");
    printf("\n");
}

int main(void) {
    char str[SIZE];
    int choice;
    do {
        printf("\n1. Reverse each word\n");
        printf("2. Print initials\n3. Exit\n");
        printf("Enter your choice: ");
        if (scanf("%d", &choice) != 1) {
            printf("Invalid input.\n");
            return 1;
        }
        while (getchar() != '\n');
        switch (choice) {
            case 1:
                printf("Enter a sentence: ");
                readLine(str, SIZE);
                reverseWords(str);
                printf("Reversed words: %s\n", str);
                break;
            case 2:
                printf("Enter a name: ");
                readLine(str, SIZE);
                printf("Initials: ");
                printInitials(str);
                break;
            case 3:
                printf("Exiting program.\n");
                break;
            default:
                printf("Invalid choice. Try again.\n");
        }
    } while (choice != 3);
    return 0;
}
```

Sample Output

```
1. Reverse each word
2. Print initials
3. Exit
Enter your choice: 1
Enter a sentence: What is special?
Reversed words: tahW si ?laiceps

1. Reverse each word
2. Print initials
3. Exit
Enter your choice: 2
Enter a name: Alexander Grahambell
Initials: A.G.

Enter your choice: 3
Exiting program.
```

Result

The menu-driven C program was written successfully. It reverses each word without changing the word order and prints the initials of a given name.